

Graded sphere: tensors and orbits of a sphere with density $\rho(1 + \varepsilon z)$

Free-molecule response tensors, steady states, righting dynamics, and interactive simulations

A sphere of radius l whose density varies linearly along one body axis, $\rho(\mathbf{r}) = \rho_p(1 + \varepsilon z)$, is geometrically still a perfect sphere. Every gas-surface stress therefore acts about its geometric centre H. The density gradient only moves the centre of mass (CoM) off H and changes the inertia tensor. This notebook derives what that implies, using the corrected surface-element stress of the notes (v22, Eq. 50) as the only physical input.

Results in brief. (1) About the CoM the particle has isotropic thermophoretic and drag tensors, plus three new blocks: an antisymmetric thermophoretic torque $\mathbf{T}_{\text{th}} = \gamma_0 [\mathbf{r}_H]_{\times}$, an antisymmetric translation-rotation drag coupling, and a parallel-axis correction to the rotational drag. Here $\mathbf{r}_H = -z_c \hat{\mathbf{e}}_3$ and $z_c = \varepsilon l^2 / 5$. This is exactly the polar class $C_{\infty v}$ of the notes' Eq. (29a), with $\lambda_a = -\gamma_0 z_c$. (2) The torque rights the particle heavy side down, with stiffness $M g z_c$, like a damped pendulum. (3) $\mathbf{T}_{\text{th}}$ has no real eigenvector with non-zero eigenvalue, so there is **no steady spin and no circular orbit**: the particle levitates and hovers. (4) The righting happens about the force centre: H moves during the swing but ends exactly where it started, while the CoM swings underneath it and ends lower by $z_c(1 - \cos \theta_0)$.

1. Setup

```

ClearAll["Global`*"];
$verifications = {};
record[ref_, claim_, status_, detail_] :=
  (AppendTo[$verifications, {ref, claim, status, detail}];
   VR[ref, claim, status, detail]);
VR /: MakeBoxes[VR[ref_, claim_, status_, detail_, fmt_] :=
  ToBoxes[Panel[Column[{Row[{Style[status, Bold,
    If[status == "VERIFIED", Darker[Green], Red]], " ", Style[ref, Bold],
    ": ", claim}], If[detail == "", Nothing, detail]}]], fmt];
check[ref_String, claim_String, test_, detail_ : "" ] :=
  record[ref, claim, If[TrueQ[test], "VERIFIED", "DISCREPANCY"], detail];
zeroQ[expr_] := AllTrue[Flatten[{Simplify[expr]}], PossibleZeroQ];
sci[x_] := If[x == 0, "0", Module[{e = Floor[Log10[Abs[N[x]]]}],
  ToString[NumberForm[N[x] / 10^e, {3, 2}]] <> "e" <> ToString[e]]];
num[x_] := ToString[NumberForm[N[x], {5, 3}]];
hat[{a1_, a2_, a3_}] := {{0, -a3, a2}, {a3, 0, -a1}, {-a2, a1, 0}};
quatR[{q0_, q1_, q2_, q3_}] := (q0^2 - {q1, q2, q3} . {q1, q2, q3}) IdentityMatrix[3] +
  2 Outer[Times, {q1, q2, q3}, {q1, q2, q3}] + 2 q0 hat[{q1, q2, q3}];
Qmat[{q0_, q1_, q2_, q3_}] :=
  {{-q1, -q2, -q3}, {q0, -q3, q2}, {q3, q0, -q1}, {-q2, q1, q0}};
Zhat = {0, 0, 1}; e3 = {0, 0, 1};
$Assumptions = {l > 0, rho0 > 0, a1 > 0, hh > 0,
  kap > 0, Tk > 0, 0 < aa <= 1, -1 < eps l < 1, Element[zc, Reals]};

```

2. Mass properties

Measure body coordinates from the geometric centre H. The density $\rho_p(1 + \varepsilon z)$ stays positive as long as $|\varepsilon| < 1$. Odd moments of the sphere vanish, so the mass is unchanged and the CoM moves along $\hat{e}_3$ by $z_c = \varepsilon \langle z^2 \rangle = \varepsilon l^2 / 5$. The inertia about H is also unchanged, $\frac{2}{5} M l^2 I_3$. About the CoM, the parallel-axis theorem lowers the two transverse moments.

```

rho[r_, th_] := rho0 (1 + eps r Cos[th]);
sphInt[f_] := Integrate[f r^2 Sin[th], {r, 0, l}, {th, 0, Pi}, {ph, 0, 2 Pi}];
Mass = sphInt[rho[r, th]];
com = sphInt[rho[r, th] r {Sin[th] Cos[ph], Sin[th] Sin[ph], Cos[th]}} / Mass;
inertiaH = sphInt[rho[r, th] (r^2 IdentityMatrix[3] - Outer[Times, #, #] &[
  r {Sin[th] Cos[ph], Sin[th] Sin[ph], Cos[th]}])];
inertiaCoM =
  Simplify[inertiaH - Mass (com.com IdentityMatrix[3] - Outer[Times, com, com])];
Column[{Row[{"M = ", TraditionalForm[Simplify[Mass]]},
  ",   CoM = ", TraditionalForm[Simplify[com]]}],
  Row[{"I(H) = ", TraditionalForm[MatrixForm[Simplify[inertiaH]]],
  "      I(CoM) = ", TraditionalForm[MatrixForm[Simplify[inertiaCoM]]}],
  check["mass", "M = (4/3) pi l^3 rho0, independent of eps",
    zeroQ[Mass - 4 / 3 Pi l^3 rho0]],
  check["CoM", "CoM displaced by z_c = eps l^2/5 along e3",
    zeroQ[com - {0, 0, eps l^2 / 5}]], check["inertia",
    "inertia about H unchanged; about CoM the transverse moments drop by M z_c^2",
    zeroQ[inertiaH - 2 / 5 Mass l^2 IdentityMatrix[3]] && zeroQ[inertiaCoM - (2 / 5 Mass
      l^2 IdentityMatrix[3] - Mass (eps l^2 / 5)^2 DiagonalMatrix[{1, 1, 0}])]]]

```

$$M = \frac{4}{3} \pi l^3 \rho_0, \quad \text{CoM} = \left\{ 0, 0, \frac{\epsilon l^2}{5} \right\}$$

$$I(H) = \begin{pmatrix} \frac{8}{15} \pi l^5 \rho_0 & 0 & 0 \\ 0 & \frac{8}{15} \pi l^5 \rho_0 & 0 \\ 0 & 0 & \frac{8}{15} \pi l^5 \rho_0 \end{pmatrix} \quad I(\text{CoM}) =$$

$$\begin{pmatrix} -\frac{4}{75} \pi l^5 \rho_0 (\epsilon^2 l^2 - 10) & 0 & 0 \\ 0 & -\frac{4}{75} \pi l^5 \rho_0 (\epsilon^2 l^2 - 10) & 0 \\ 0 & 0 & \frac{8}{15} \pi l^5 \rho_0 \end{pmatrix}$$

✓ **VERIFIED mass:** $M = (4/3) \pi l^3 \rho_0$, independent of ϵ

✓ **VERIFIED CoM:** CoM displaced by $z_c = \epsilon l^2/5$ along e_3

✓ **VERIFIED inertia:** inertia about H unchanged; about CoM the transverse moments drop by $M z_c^2$

3. Response tensors from the surface stresses

The only physical input is the free-molecule stress on a surface element (notes v22, Eq. 50). It applies to a convex body, where every element sees the undisturbed gas. Here $\mathbf{V}$ is the wall velocity relative to the gas, $\mathbf{q} = -\kappa T \nabla \ln T$ is the heat flux, a is the accommodation coefficient, $\alpha = m N / \sqrt{\pi h}$ and

$$h = m/2 k_B T:$$

$$\boldsymbol{\sigma} = -\alpha \left[\frac{a}{2} \mathbf{V}_t + \left(2 - a + \frac{\pi a}{4} \right) (\mathbf{V} \cdot \hat{n}) \hat{n} \right] + \frac{a \sqrt{h}}{5 \sqrt{\pi}} \mathbf{q}_t + \frac{2(2-a) \sqrt{h}}{5 \sqrt{\pi}} (\mathbf{q} \cdot \hat{n}) \hat{n} \quad (\text{pressure omitted : it cancels on a closed surface})$$

Put the CoM at the origin. The sphere's surface is then $\mathbf{r} = \mathbf{r}_H + l \hat{n}$ with $\mathbf{r}_H = -z_c \hat{e}_3$. For a body moving with velocity $\mathbf{v}$ and angular velocity $\boldsymbol{\omega}$ (body frame), the wall velocity is $\mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}$. Integrating $\boldsymbol{\sigma}$ and $\mathbf{r} \times \boldsymbol{\sigma}$ over the sphere gives the complete linear response about the CoM, $(\mathbf{F}, \boldsymbol{\tau}) = -\mathcal{R} \cdot (\nabla \ln T, \mathbf{v}, \boldsymbol{\omega})$, a 6×9 matrix.

```

nH[th_, ph_] := {Sin[th] Cos[ph], Sin[th] Sin[ph], Cos[th]};
stress[n_, V_, q_] := -al ((aa/2) (V - (V.n) n) + (2 - aa (1 - Pi/4)) (V.n) n) +
  aa Sqrt[hh] / (5 Sqrt[Pi]) (q - (q.n) n) + 2 (2 - aa) Sqrt[hh] / (5 Sqrt[Pi]) (q.n) n;
rH = {0, 0, -zc}; vv = {v1, v2, v3}; ww = {w1, w2, w3}; gg = {g1, g2, g3};
wrench = Simplify[Integrate[Module[{n = nH[th, ph], r, s}, r = rH + l n;
  s = stress[n, vv + Cross[ww, r], -kap Tk gg];
  Join[s, Cross[r, s]] l^2 Sin[th]], {th, 0, Pi}, {ph, 0, 2 Pi}]]];
Rsp = -D[wrench, {Join[gg, vv, ww]}];
gamma0 = Rsp[[1, 1]]; xi0 = Rsp[[1, 4]]; c0 = Rsp[[6, 9]];
Column[{Row[{"γ0 = ", TraditionalForm[gamma0], ",   ξ0 = ",
  TraditionalForm[Factor[xi0]], ",   c0 = ", TraditionalForm[c0]}],
  Row[{"ℛ = ", TraditionalForm[MatrixForm[Rsp /. {gamma0 → "γ0"}]}]}]
γ0 = 16/15 √π √hh kap l^2 Tk,   ξ0 = 1/3 π (π aa + 8) al l^2,   c0 = 4/3 π aa al l^4
ℛ = ⎛⎜⎝
      γ0      0      0      1/3 π (π aa + 8) al l^2      0
      0      γ0      0      0      1/3 π (π aa + 8) al l^2
      0      0      γ0      0      0
      0      γ0 zc  0      0      1/15 (5 π^2 aa al l^2 zc + 40 π al l
-16/15 √π √hh kap l^2 Tk zc  0      0  -1/3 π (π aa + 8) al l^2 zc      0
      0      0      0      0      0

```

Read the blocks off, in the notation of the notes plus the coupling block $\mathbf{B}$ that the notes omit:

$$\mathbf{f}_{\text{th}} = \gamma_0 \mathbf{I}_3, \quad \mathbf{f}_{\text{drag}} = \xi_0 \mathbf{I}_3, \quad \mathbf{T}_{\text{th}} = \gamma_0 [\mathbf{r}_H]_x = -\gamma_0 z_c [\hat{e}_3]_x, \quad \mathbf{T}_{\text{drag}} = c_0 \mathbf{I}_3 + \xi_0 z_c^2 \text{diag}(1, 1, 0), \quad \mathbf{B} = -\xi_0 [\mathbf{r}_H]_x$$

$$\gamma_0 = \frac{16 \sqrt{\pi}}{15} \frac{\kappa T l^2}{\bar{v}}, \quad \xi_0 = \frac{4 \pi}{3} l^2 \rho_g \bar{c} \left(1 + \frac{\pi a}{8} \right), \quad c_0 = \frac{2 \pi}{3} a \rho_g \bar{c} l^4, \quad \bar{v} = \sqrt{2 k_B T / m} = 1 / \sqrt{h}, \quad \rho_g \bar{c} = 2 \alpha$$

Everything about the CoM follows from forces applied at H. The thermophoretic and drag forces act at H, so they exert torques $\mathbf{r}_H \times \mathbf{F}$ about the CoM, and rotation about the CoM moves H, so it drags. The 6×6 resistance block $\begin{pmatrix} \mathbf{f}_{\text{drag}} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{T}_{\text{drag}} \end{pmatrix}$ is symmetric (Onsager) and positive-definite.

```
Tth = Rsp[[4 ;; 6, 1 ;; 3]];
Bblk = Rsp[[1 ;; 3, 7 ;; 9]];
Res6 = Rsp[All, 4 ;; 9];
Column[{check["Waldmann", "gamma0 = (16 Sqrt[pi]/15) kappa T l^2/vbar",
  zeroQ[gamma0 - 16 Sqrt[Pi] / 15 kap Tk l^2 Sqrt[hh]]], check["Epstein",
  "xi0 = (4 pi/3) l^2 (rho cbar) (1 + pi a/8) with rho cbar = 2 alpha",
  zeroQ[xi0 - 4 Pi / 3 l^2 (2 al) (1 + Pi aa / 8)]], check["rotation",
  "c0 = (2 pi/3) a (rho cbar) l^4", zeroQ[c0 - 2 Pi / 3 aa (2 al) l^4]],
check["T_th", "T_th = gamma0 [r_H]x (antisymmetric, lambda_a = -gamma0 z_c)",
  zeroQ[Tth - gamma0 hat[rH]]],
check["coupling", "B = -xi0 [r_H]x and T_drag = c0 + xi0 z_c^2 diag(1,1,0)",
  zeroQ[Bblk + xi0 hat[rH]] && zeroQ[
    Rsp[[4 ;; 6, 7 ;; 9] - (c0 IdentityMatrix[3] + xi0 zc^2 DiagonalMatrix[{1, 1, 0}])]],
check["Onsager", "6x6 resistance matrix is symmetric",
  zeroQ[Res6 - Transpose[Res6]]], check["shift rule",
  "all blocks follow from the sphere tensors at H by the rigid shift r_H",
Module[
  {P = ArrayFlatten[{{IdentityMatrix[3], -hat[rH]}, {0, IdentityMatrix[3]}}],
  thH, resH}, thH = Join[gamma0 IdentityMatrix[3], ConstantArray[0, {3, 3}]];
  resH = ArrayFlatten[{{xi0 IdentityMatrix[3], 0}, {0, c0 IdentityMatrix[3]}}];
  zeroQ[Rsp - Join[Transpose[P].thH, Transpose[P].resH.P, 2]]]]]
```

✓ **VERIFIED** **Waldmann:** $\gamma_0 = (16 \sqrt{\pi}/15) \kappa T l^2/v_{\text{bar}}$

✓ **VERIFIED** **Epstein:** $\xi_0 = (4 \pi/3) l^2 (\rho \text{ cbar})(1 + \pi a/8)$ with $\rho \text{ cbar} = 2 \alpha$

✓ **VERIFIED** **rotation:** $c_0 = (2 \pi/3) a (\rho \text{ cbar}) l^4$

✓ **VERIFIED** **T_th:** $T_{\text{th}} = \gamma_0 [r_H]x$ (antisymmetric, $\lambda_a = -\gamma_0 z_c$)

✓ **VERIFIED** **coupling:** $B = -\xi_0 [r_H]x$ and $T_{\text{drag}} = c_0 + \xi_0 z_c^2 \text{diag}(1,1,0)$

✓ **VERIFIED** **Onsager:** 6x6 resistance matrix is symmetric

✓ **VERIFIED** **shift rule:** all blocks follow from the sphere tensors at H by the rigid shift r_H

4. Steady states: levitation, orientation, and the absence of orbits

Take the uniform gradient of the notes, $\nabla \ln T = -G \hat{Z}$. Because $\mathbf{f}_{\text{th}}$ is isotropic, the thermophoretic force is $\gamma_0 G \hat{Z}$ in every orientation, so the levitation gradient is exactly $G_0 = Mg/\gamma_0$, independent of ϵ . The force acts at H while gravity acts at the CoM. The torque $\mathbf{r}_H \times \gamma_0 G \hat{Z}$ vanishes only when $\mathbf{r}_H$ is vertical. The stable orientation has H above the CoM, i.e. the heavy side down, with restoring stiffness $\kappa_\theta = Mg z_c$ at $G = G_0$.

In the notes' language (Section 2.2), $\delta \mathbf{T}_{\text{th}}$ is antisymmetric. Its only real eigenvector is $\hat{\mathbf{e}}_3$, with eigenvalue $\lambda = 0$, so the steady state has $\mathbf{\Omega} = 0$: **no spin, hence no precession and no circular orbit** (Eq. 49 gives nothing to divide by). The particle hovers. The notes' symmetric-eigenvector analysis does not apply to an antisymmetric $\delta \mathbf{T}_{\text{th}}$: the torque here does not spin the body, it *aligns* it.

```
Module[{gl = -G Zhat, torqueAtTilt, eig}, torqueAtTilt[th_] :=
  Module[{R = RotationMatrix[th, {0, 1, 0}].RotationMatrix[Pi, {1, 0, 0}]},
    R.(-Tth.(Transpose[R].gl));
  eig = Eigensystem[Tth /. zc -> 1];
  Column[{Row[{"torque (lab) at heavy-axis tilt  $\theta$ :"}, TraditionalForm[
    Simplify[torqueAtTilt[ $\theta$ ] /. G -> Mg / gamma0]]], check["levitation",
    "levitation gradient  $G_0 = M g/\gamma_0$  is independent of  $\epsilon$  ( $f_{\text{th}}$  isotropic)",
    zeroQ[Rsp[[1 ;; 3, 1 ;; 3]] - gamma0 IdentityMatrix[3]]], check["righting",
    "restoring torque  $-M g z_c \sin(\theta)$  about the horizontal axis at  $G = G_0$ ",
    zeroQ[(torqueAtTilt[ $\theta$ ] /. G -> Mg / gamma0) - {0, -Mg zc Sin[ $\theta$ ], 0}]],
    check["no spin", "the only real eigenvector of  $T_{\text{th}}$  is  $\mathbf{e}_3$ , with eigenvalue 0:
    no steady spin, no orbit", Select[eig[[1]], Element[#, Reals] &] === {0}]]]]
torque (lab) at heavy-axis tilt  $\theta$ : {0, -Mg zc sin( $\theta$ ), 0}
```

✓ **VERIFIED levitation:** levitation gradient $G_0 = M g/\gamma_0$ is independent of ϵ (f_{th} isotropic)

✓ **VERIFIED righting:** restoring torque $-M g z_c \sin(\theta)$ about the horizontal axis at $G = G_0$

✓ **VERIFIED no spin:** the only real eigenvector of T_{th} is $\mathbf{e}_3$, with eigenvalue 0: no steady spin, no orbit

5. Linear dynamics: a damped pendulum that rights about H

Tilt the heavy axis by a small angle θ about $\hat{Y}$ from straight down at $G = G_0$, and let the CoM move horizontally by X . The force centre is at $X_H = X + z_c \theta$ and its velocity drags. The horizontal force and the torque are

$$M\ddot{X} = -\xi_0(\dot{X} + z_c \dot{\theta}), \quad I_\perp \ddot{\theta} = -z_c \xi_0(\dot{X} + z_c \dot{\theta}) - c_0 \dot{\theta} - M g z_c \theta,$$

with $I_\perp = \frac{2}{5} M l^2 - M z_c^2$. The coupling is symmetric, as reciprocity requires. The characteristic polynomial, after removing the neutral translation mode $s = 0$, is a cubic in which the coupling cancels from the linear coefficient:

$$M I_\perp s^3 + (M c_0 + M z_c^2 \xi_0 + \xi_0 I_\perp) s^2 + (M^2 g z_c + \xi_0 c_0) s + \xi_0 M g z_c = 0.$$

Integrating the X equation once gives $M \dot{X} + \xi_0(X + z_c \theta) = \text{const}$. When the particle comes to rest, $X_H = X + z_c \theta$ has returned to its initial value: **H ends where it started; the CoM swings underneath it**. The result is exact, not just linear. Because $\mathbf{f}_{\text{th}}$ and $\mathbf{f}_{\text{drag}}$ are isotropic, the net force is $-\xi_0 \dot{\mathbf{x}}_H$ plus the balanced weight and thermophoresis, so $M \dot{\mathbf{v}} = -\xi_0 \dot{\mathbf{x}}_H$ integrates to $\mathbf{x}_H(\infty) = \mathbf{x}_H(0) + M \mathbf{v}(0)/\xi_0$. Section 6 checks this numerically.

```
lin = Module[{Ip = 2 / 5 Mm l^2 - Mm zc^2, mat, poly},
  mat = {{Mm s^2 + xi0 s, xi0 zc s}, {xi0 zc s, Ip s^2 + (c0 + xi0 zc^2) s + Mm gg0 zc}};
  poly = Factor[Det[mat]];
  {poly, Collect[Simplify[poly / s], s, Simplify]};
Column[{Row[{"det = ", TraditionalForm[lin[[1]]}], check["cubic",
  "linear coefficient is M^2 g z_c + xi0 c0 (the coupling cancels)",
  zeroQ[Coefficient[Expand[lin[[2]]], s, 1] - (Mm^2 gg0 zc + xi0 c0)]]],
det =
  1
  --- s (20 \pi^3 aa^2 al^2 l^6 s + 160 \pi^2 aa al^2 l^6 s + 15 \pi^2 aa al gg0 l^2 Mm zc + 6 \pi^2 aa al l^4 Mm s^2 +
  45
    60 \pi aa al l^4 Mm s^2 + 120 \pi al gg0 l^2 Mm zc + 48 \pi al l^4 Mm s^2 +
    45 gg0 Mm^2 s zc + 18 l^2 Mm^2 s^3 - 45 Mm^2 s^3 zc^2)
```

✓ **VERIFIED cubic:** linear coefficient is $M^2 g z_c + \xi_0 c_0$ (the coupling cancels)

6. Realistic particles

The defaults follow the notes' Section 6: $l = 20 \mu\text{m}$, $\rho_p = 1 \text{ g/cm}^3$, air at $T = 300 \text{ K}$ and $P = 5 \text{ Torr}$, full accommodation, and $\kappa = 0.026 \text{ W/(m K)}$, with $\varepsilon l = 0.5$. The temperature field is the notes' uniform gradient $\nabla \ln T = -G \hat{Z}$ with $G = f_G G_0$. An optional harmonic curvature (radial and vertical trap frequencies f_r, f_z) represents the confinement of the real cell. At $K n \approx 0.5$ the free-molecule coefficients are only indicative.

```

$kB = 1.380649*^-23; $mAir = 4.81*^-26; $g = 9.81;
physics[pars_Association] :=
Module[{p = Join[<|"radius" → 20*^-6, "rhoP" → 1000., "epsL" → 0.5, "T" → 300.,
  "PTorr" → 5., "kappa" → 0.026, "a" → 1., "fG" → 1., "fr" → 0., "fz" → 0.|>, pars],
  l0, N0, h0, al0, M0, zc0, subs, gam, xi, cr, Ip, I3, mu, lam, G0}, l0 = p["radius"];
N0 = p["PTorr"] 133.322 / ($kB p["T"]);
h0 = $mAir / (2 $kB p["T"]);
al0 = $mAir N0 / Sqrt[Pi h0];
M0 = 4 / 3 Pi l0^3 p["rhoP"];
zc0 = p["epsL"] l0 / 5;
subs =
  {l → l0, al → al0, hh → h0, kap → p["kappa"], Tk → p["T"], aa → p["a"], zc → zc0};
{gam, xi, cr} = {gamma0, xi0, c0} /. subs;
G0 = M0 $g / gam;
Ip = 2 / 5 M0 l0^2 - M0 zc0^2;
I3 = 2 / 5 M0 l0^2;
mu = 1.716*^-5 (p["T"] / 273.15) ^1.5 (273.15 + 110.4) / (p["T"] + 110.4);
lam = mu / (p["PTorr"] 133.322) Sqrt[Pi $kB p["T"] / (2 $mAir)];
Join[p, <|"M" → M0, "zc" → zc0, "R" → N[Rsp /. subs], "gamma0" → gam,
  "xi0" → xi, "c0" → cr, "I" → DiagonalMatrix[{Ip, Ip, I3}], "G0" → G0,
  "gradT0 (K/cm)" → G0 p["T"] / 100, "Kn" → lam / l0, "tauV" → M0 / xi,
  "tauR" → Ip / cr, "A" → DiagonalMatrix[{M0 (2 Pi p["fr"]) ^2 / gam,
    M0 (2 Pi p["fr"]) ^2 / gam, M0 (2 Pi p["fz"]) ^2 / gam}]|>]];
linearModes[ph_] :=
Module[{s, roots}, roots = s /. NSolve[ph["M"] ph["I"] [[1, 1]] s^3 + (ph["M"] ph["c0"] +
  ph["M"] ph["zc"] ^2 ph["xi0"] + ph["xi0"] ph["I"] [[1, 1]] s^2 +
  (ph["M"] ^2 $g Abs[ph["zc"]] + ph["xi0"] ph["c0"]) s +
  ph["xi0"] ph["M"] $g Abs[ph["zc"]] = 0, s]; <|
  "roots" → roots, "pendulum f (Hz)" → Max[Abs[Im[roots]]] / (2 Pi),
  "pendulum decay (1/s)" → -Re[First[SortBy[roots, -Abs[Im[#]] &]]]|>];
ph0 = physics[<|>];
Grid[{{"M (kg)", ph0["M"]},
  {"z_c (um)", 10^6 ph0["zc"]}, {"gamma0 (N m)", ph0["gamma0"]},
  {"xi0 (N s/m)", ph0["xi0"]}, {"c0 (N m s)", ph0["c0"]},
  {"levitation |grad T| (K/cm)", ph0["gradT0 (K/cm)"]}, {"Kn", ph0["Kn"]},
  {"M/xi0 (ms)", 1000 ph0["tauV"]}, {"I_perp/c0 (ms)", 1000 ph0["tauR"]},
  {"pendulum frequency (Hz)", linearModes[ph0]["pendulum f (Hz)"]},
  {"pendulum decay rate (1/s)", linearModes[ph0]["pendulum decay (1/s)"]}},
  Frame → All, Alignment → Left]

```

M (kg)	3.35103×10^{-11}
z_c (um)	2.
gamma0 (N m)	1.42139×10^{-11}
xi0 (N s/m)	8.45898×10^{-9}
c0 (N m s)	1.21476×10^{-18}
levitation grad T (K/cm)	69.3834
Kn	0.509213
M/xi0 (ms)	3.96151
I_perp/c0 (ms)	4.30341
pendulum frequency (Hz)	52.8596
pendulum decay rate (1/s)	117.818

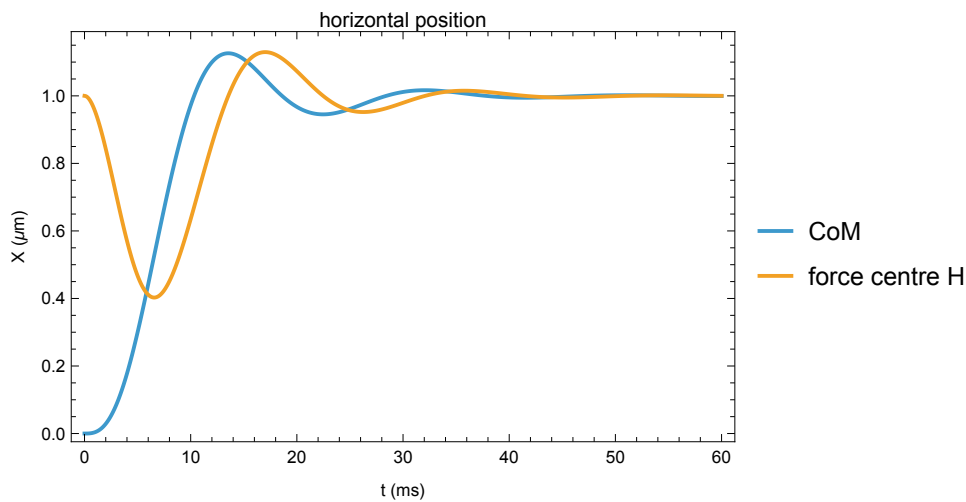
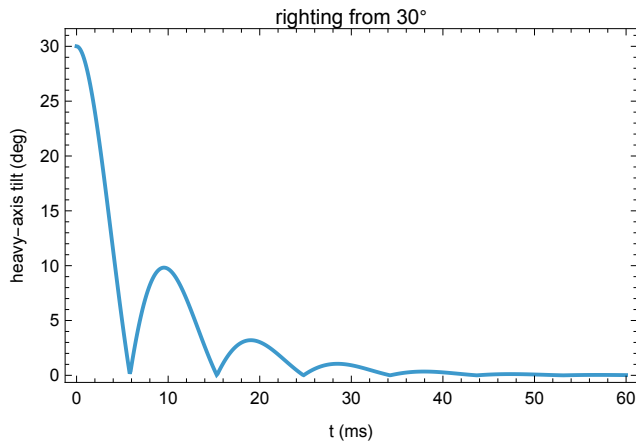
Simulator. This integrates the full nonlinear rigid-body equations (notes, Eqs. 23–26), extended by the coupling blocks: body-frame wrench $-\mathcal{R} \cdot (R^T \nabla \ln T(\mathbf{x}_H), R^T \mathbf{v}, \boldsymbol{\omega})$, gravity at the CoM, and the field evaluated at the force centre $\mathbf{x}_H = \mathbf{x} + R \mathbf{r}_H$. Lengths are scaled by l and times by 1 ms internally.

```
simulate[ph_, tiltDeg_, spin_, tEndMs_, x0_ : {0, 0, 0}] :=
Module[{M0 = ph["M"], I0 = ph["I"], Iinv, Rm = ph["R"],
  rHv = {0, 0, -ph["zc"]}, A = ph["A"], G = ph["fG"] ph["G0"], l0 = ph["radius"],
  t0 = 10^-3, rhs, f, y, t, sol, qDown, q0, sc}, Iinv = Inverse[I0];
qDown = If[ph["epsL"] >= 0, {0., 1., 0., 0.}, {1., 0., 0., 0.}];
q0 = Normalize[{Cos[tiltDeg Degree / 2], 0, Sin[tiltDeg Degree / 2], 0}];
q0 = Module[{a = q0, b = qDown}, Prepend[a[[1]] b[[2 ;; 4]] + b[[1]] a[[2 ;; 4]] +
  Cross[a[[2 ;; 4]], b[[2 ;; 4]], a[[1]] b[[1]] - a[[2 ;; 4]].b[[2 ;; 4]]];
rhs[s_] := Module[{x = s[[1 ;; 3]], v = s[[4 ;; 6]], q = s[[7 ;; 10]],
  w = s[[11 ;; 13]], R, wr}, R = quatR[q / Sqrt[q.q]];
wr = -Rm.Join[Transpose[R].(A.(x + R.rHv) - G.Zhat), Transpose[R].v, w];
Join[v, R.wr[[1 ;; 3]] / M0 - $g.Zhat,
  1 / 2 Qmat[q].w, Iinv.(wr[[4 ;; 6]] - Cross[w, I0.w])];
sc = Join[ConstantArray[l0, 3], ConstantArray[l0 / t0, 3],
  {1, 1, 1, 1}, ConstantArray[1 / t0, 3]];
f[yy_? (VectorQ[#] & NumericQ)] := t0 rhs[sc yy] / sc;
sol = NDSolveValue[{y'[t] == f[y[t]], y[0] == Join[x0 / l0, {0, 0, 0}, q0, spin t0]}, y,
  {t, 0, tEndMs}, MaxSteps -> 10^6, PrecisionGoal -> 9, AccuracyGoal -> 10]; <|
"ph" -> ph, "tEnd" -> tEndMs, "state" -> Function[{tm},
  Module[{yy = sol[tm]}, Join[l0 yy[[1 ;; 6]] {1, 1, 1, 1 / t0, 1 / t0, 1 / t0},
    Normalize[yy[[7 ;; 10]], yy[[11 ;; 13]] / t0]]]; >|
heavyTilt[q_, ph_] :=
ArcCos[Clip[-(quatR[q].(Sign[ph["epsL"]] e3)).Zhat, {-1, 1}]] / Degree;
forceCentre[st_, ph_] := st[[1 ;; 3]] + quatR[st[[7 ;; 10]].{0, 0, -ph["zc"]];
```

```

run0 = simulate[ph0, 30, {0, 0, 0}, 150];
Module[{ts = Range[0, 60, 0.1], st, tilt, X, XH, lm = linearModes[ph0], fit},
  st = run0["state"] /@ ts;
  tilt = heavyTilt[#[[7 ;; 10]], ph0] & /@ st;
  X = 10^6 st[[All, 1]];
  XH = 10^6 (forceCentre[#, ph0][[1]] & /@ st);
  Column[{Row[{ListLinePlot[Transpose[{ts, tilt}],
    Frame → True, FrameLabel → {"t (ms)", "heavy-axis tilt (deg)"},
    PlotLabel → "righting from 30°", ImageSize → 330, PlotRange → All],
    ListLinePlot[{Transpose[{ts, X}], Transpose[{ts, XH}]], Frame → True,
    FrameLabel → {"t (ms)", "X (μm)"}, PlotLegends → {"CoM", "force centre H"},
    PlotLabel → "horizontal position", ImageSize → 380, PlotRange → All}]], check[
    "nonlinear", "H returns to its start: X_H(end) = X_H(0) = z_c sin(theta0)",
    Abs[XH[[-1]] - 10^6 ph0["zc"] Sin[30 Degree]] < 10^-3 Abs[XH[[1]]],
    "X_H(0) = "<> num[XH[[1]]]<> " um, X_H(end) = "<> num[XH[[-1]]]<>
    " um; the CoM moved from 0 to "<> num[X[[-1]]]<> " um."],
  check["frequency", "small-tilt oscillation frequency matches the linear cubic",
    Module[{run = simulate[ph0, 2, {0, 0, 0}, 40], tt = Range[0, 40, 0.01], th, zeros},
      th = (quatR[run["state"]][#[[7 ;; 10]].(Sign[ph0["epsL"]][e3])[1]] & /@ tt;
      zeros = Select[Partition[Transpose[{tt, th}], 2, 1],
        #[[1, 2]]#[[2, 2]] < 0 &][[All, 1, 1]];
      Abs[1000 / (2 Mean[Differences[zeros]]) / lm["pendulum f (Hz)"] - 1] < 0.02],
    "linear prediction "<> ToString[NumberForm[lm["pendulum f (Hz)"], 4]]<> " Hz"],
  check["steady state",
    "after 150 ms the particle is at rest: no spin and no orbit",
    Norm[run0["state"]][150][[11 ;; 13]] < 10^-3 && Norm[run0["state"]][150][[4 ;; 6]] <
      10^-9, "|omega| = "<> sci[Norm[run0["state"]][150][[11 ;; 13]]]<>
      " rad/s, |v| = "<> sci[Norm[run0["state"]][150][[4 ;; 6]]]<> " m/s"]]]]

```



✓ **VERIFIED nonlinear:** H returns to its start: $X_H(\text{end}) = X_H(0) = z_c \sin(\theta_0)$

$X_H(0) = 1.000 \mu\text{m}$, $X_H(\text{end}) = 1.000 \mu\text{m}$; the CoM moved from 0 to 0.999 μm.

✓ **VERIFIED frequency:** small-tilt oscillation frequency matches the linear cubic

linear prediction 52.86 Hz

✓ **VERIFIED steady state:** after 150 ms the particle is at rest: no spin and no orbit

$|\omega| = 2.07\text{e-}6 \text{ rad/s}$, $|v| = 6.26\text{e-}12 \text{ m/s}$

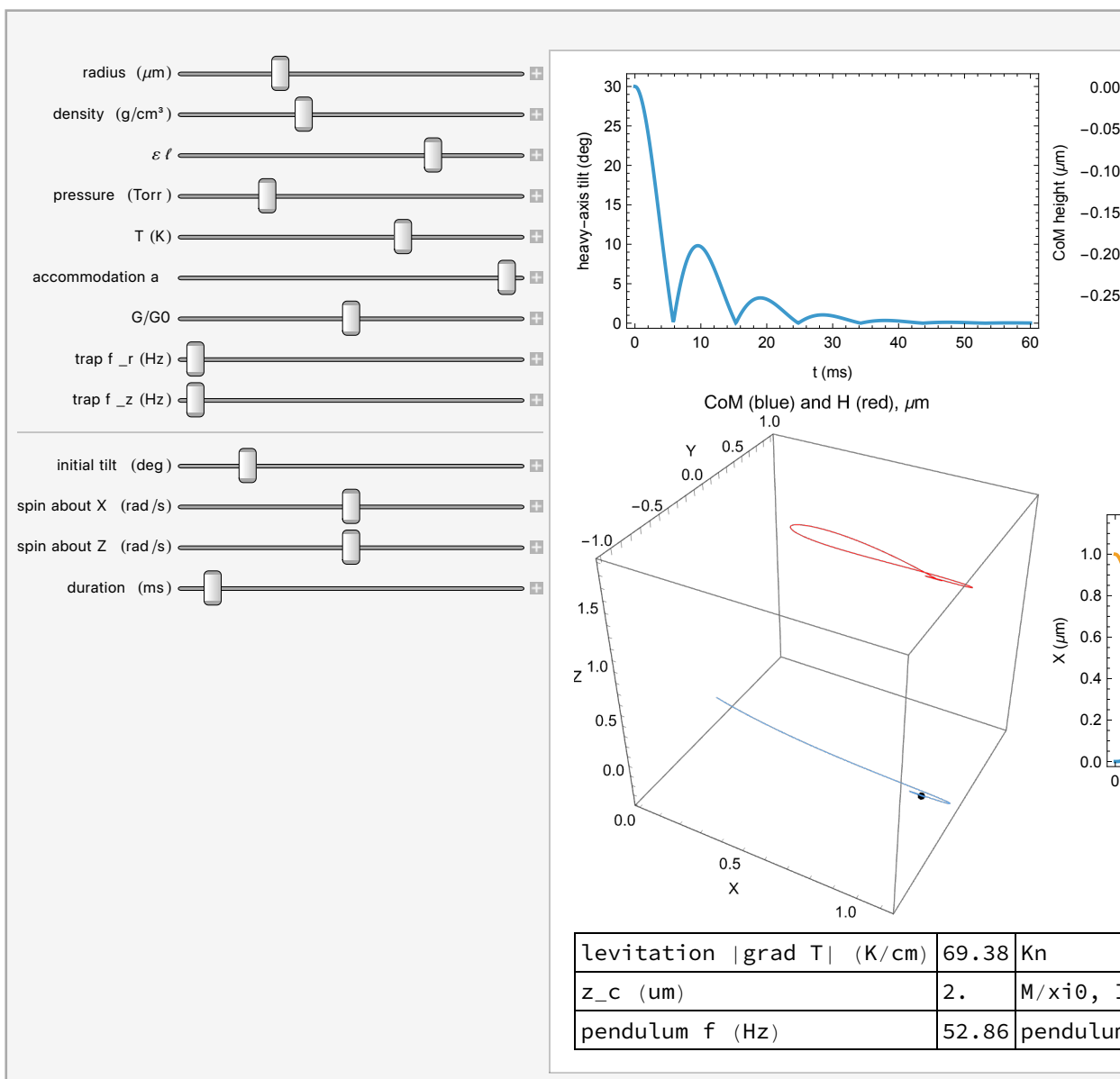
7. Interactive simulations

Explorer. Change the particle and cell parameters and the initial condition, and the notebook integrates the full equations again. Useful things to try: an initial spin about a horizontal axis, which precesses and damps; a spin about $\hat{e}_3$, which is neutral apart from drag; a gradient off levitation ($f_G \neq 1$), which gives drift at terminal velocity; a radial trap; and the inverted start (tilt near 180°), which is unstable. Each change re-runs the integration, which takes a second or two.

```

Manipulate[
Module[{ph = physics[<|"radius" → rad 10-6, "rhoP" → rhoc 1000, "epsL" → epsL,
  "PTorr" → pT, "T" → temp, "a" → acc, "fG" → fG, "fr" → fr, "fz" → fz|>,
  run, ts, st, lm}, run = simulate[ph, tilt0, {spinX, 0, spinZ}, tEnd];
ts = Subdivide[0, tEnd, 600];
st = run["state"] /@ ts;
lm = linearModes[ph];
Column[{Grid[{{ListLinePlot[Transpose[{ts, heavyTilt[#][7 ;; 10], ph] & /@ st}],
  Frame → True, FrameLabel → {"t (ms)", "heavy-axis tilt (deg)"},
  ImageSize → 270, PlotRange → All],
  ListLinePlot[{Transpose[{ts, 106 st[[All, 3]]}], Frame → True, FrameLabel →
    {"t (ms)", "CoM height (μm)"}, ImageSize → 270, PlotRange → All}],
  Graphics3D[{{RGBColor[0.3, 0.6, 0.9], Line[106 st[[All, 1 ;; 3]]},
    {Red, Line[106 (forceCentre[#], ph] & /@ st)}], {Black, PointSize[Medium],
    Point[106 st[[1, 1 ;; 3]]}}, Axes → True, AxesLabel → {"X", "Y", "Z"},
  BoxRatios → {1, 1, 1}, PlotLabel → "CoM (blue) and H (red), μm",
  ImageSize → 270], ListLinePlot[{Transpose[{ts, 106 st[[All, 1]]}],
  Transpose[{ts, 106 (forceCentre[#], ph)[1] & /@ st}]], Frame → True,
  FrameLabel → {"t (ms)", "X (μm)"}, PlotLabel → "CoM (blue), H (orange)",
  ImageSize → 270, PlotRange → All]]], Grid[{"levitation |grad T| (K/cm)",
  NumberForm[ph["gradT0 (K/cm)"], 4], "Kn", NumberForm[ph["Kn"], 3]},
  {"z_c (um)", NumberForm[106 ph["zc"], 3], "M/xi0, I/c0 (ms)",
  Row[{NumberForm[1000 ph["tauV"], 3], ", ", NumberForm[1000 ph["tauR"], 3]}]},
  {"pendulum f (Hz)", NumberForm[lm["pendulum f (Hz)"], 4],
  "pendulum decay (1/s)", NumberForm[lm["pendulum decay (1/s)"], 4]}],
  Frame → All, Alignment → Left]]], {{rad, 20., "radius (μm)"}, 5., 60.},
{{rhoc, 1., "density (g/cm3)"}, 0.2, 2.5}, {{epsL, 0.5, "ε /"}, -0.95, 0.95},
{{pT, 5., "pressure (Torr)"}, 0.5, 20.}, {{temp, 300., "T (K)"}, 200., 350.},
{{acc, 1., "accommodation a"}, 0.5, 1.}, {{fG, 1., "G/G0"}, 0.9, 1.1},
{{fr, 0., "trap f_r (Hz)"}, 0., 10.}, {{fz, 0., "trap f_z (Hz)"}, 0., 10.},
Delimiter, {{tilt0, 30., "initial tilt (deg)"}, 0., 179.}, {{spinX, 0.,
  "spin about X (rad/s)"}, -2000., 2000.}, {{spinZ, 0., "spin about Z (rad/s)"},
  -2000., 2000.}, {{tEnd, 60., "duration (ms)"}, 5., 1000.}, ContinuousAction → False,
SynchronousUpdating → False, SaveDefinitions → True, ControlPlacement → Left]

```

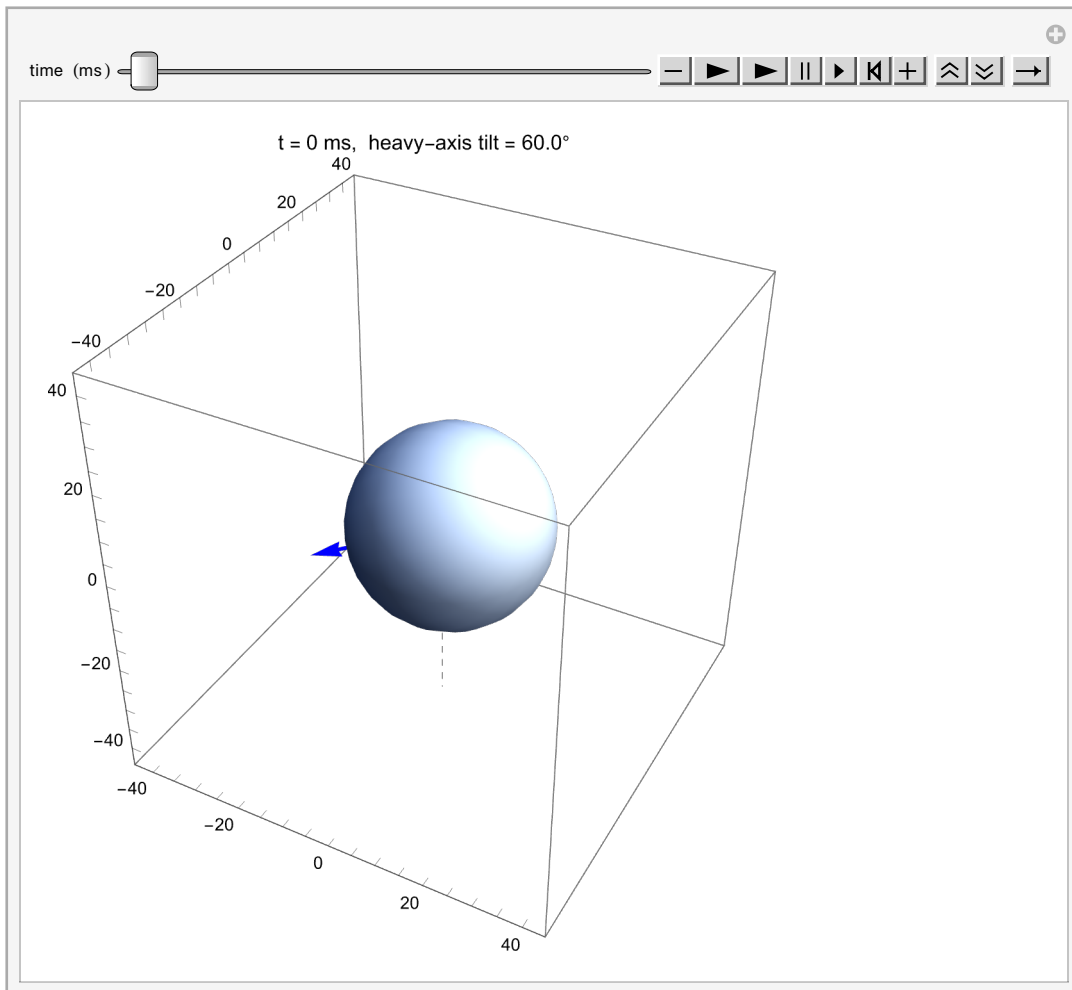


Animation. A particle rights itself from 60° , drawn with its density gradient (dark = heavy), body axis, CoM (black) and force centre H (red). Drag the time slider or press play.

```

anim = With[{ph = ph0, run = simulate[ph0, 60, {0, 0, 0}, 50], l0 = ph0["radius"]},
  Manipulate[Module[{st = run["state"][tm], R, x, sphere}, R = quatR[st[[7 ;; 10]]];
    x = 10^6 st[[1 ;; 3]];
    sphere = ParametricPlot3D[10^6
      (st[[1 ;; 3]] + R.({0, 0, -ph["zc"]} + l0 {Sin[u] Cos[v], Sin[u] Sin[v], Cos[u]})),
      {u, 0, Pi}, {v, 0, 2 Pi}, Mesh → None, PlotPoints → 25, ColorFunctionScaling →
      False, ColorFunction → Function[{xx, yy, zz, uu, vv}, Blend[{RGBColor[0.85,
        0.92, 1], RGBColor[0.1, 0.25, 0.6]}, (1 + Sign[ph["epsL"]] Cos[uu]) / 2]]],
    Show[sphere, Graphics3D[{{Black, Sphere[x, 1.2]}, {Red, Sphere[10^6 forceCentre[
      st, ph], 1.2]}, {Thick, Blue, Arrow[{x, x + 35 R. (Sign[ph["epsL"]] e3)}]},
      {Gray, Dashed, Line[{x, x - {0, 0, 40}}]}]], PlotRange →
      {{-40, 40}, {-40, 40}, {-40, 40}}, BoxRatios → {1, 1, 1}, Lighting → "Neutral",
      PlotLabel → Row[{"t = ", NumberForm[tm, {4, 1}], " ms, heavy-axis tilt = ",
        NumberForm[heavyTilt[st[[7 ;; 10]], ph], {3, 1}], "°"}], ImageSize → 380]],
    {{tm, 0, "time (ms)"}, 0, 50, 0.25, ControlType → Animator,
      AnimationRunning → False, AppearanceElements → All}, SaveDefinitions → True]]

```



8. Summary

```
Column[{Row[{"checks: ", Length[$verifications], "  ("},
  Count[$verifications[[All, 3]], "VERIFIED"], " verified)"}], Grid[Prepend[
  $verifications[[All, 1 ;; 3]], Style[#, Bold] & /@ {"item", "statement", "result"}],
  Frame → All, Alignment → Left, Background → {None, {LightGray}}]]]
checks: 17  (17 verified)
```

item	statement	result
mass	$M = (4/3) \pi l^3 \rho_0$, independent of ϵ s	VERIFIED
CoM	CoM displaced by $z_c = \epsilon l^2/5$ along e_3	VERIFIED
inertia	inertia about H unchanged; about CoM the transverse moments drop by $M z_c^2$	VERIFIED
Waldmann	$\gamma_0 = (16 \sqrt{\pi}/15) \kappa T l^2/\bar{v}$	VERIFIED
Epstein	$\xi_0 = (4 \pi/3) l^2 (\rho_0 \bar{c}) (1 + \pi a/8)$ with $\rho_0 \bar{c} = 2 \alpha$	VERIFIED
rotation	$c_0 = (2 \pi/3) a (\rho_0 \bar{c}) l^4$	VERIFIED
T_th	$T_{th} = \gamma_0 [r_H]_x$ (antisymmetric, $\lambda_a = -\gamma_0 z_c$)	VERIFIED
coupling	$B = -\xi_0 [r_H]_x$ and $T_{drag} = c_0 + \xi_0 z_c^2 \text{diag}(1,1,0)$	VERIFIED
Onsager	6x6 resistance matrix is symmetric	VERIFIED
shift rule	all blocks follow from the sphere tensors at H by the rigid shift r_H	VERIFIED
levitation	levitation gradient $G_0 = M g/\gamma_0$ is independent of ϵ s (f_{th} isotropic)	VERIFIED
righting	restoring torque $-M g z_c \sin(\theta)$ about the horizontal axis at $G = G_0$	VERIFIED
no spin	the only real eigenvector of T_{th} is e_3 , with eigenvalue 0: no steady spin, no orbit	VERIFIED
cubic	linear coefficient is M^2 $g z_c + \xi_0 c_0$ (the coupling cancels)	VERIFIED
nonlinear	H returns to its start: $X_H(\text{end}) = X_H(0) = z_c \sin(\theta_0)$	VERIFIED
frequency	small-tilt oscillation frequency matches the linear cubic	VERIFIED
steady state	after 150 ms the particle is at rest: no spin and no orbit	VERIFIED

- **Tensors.** A density-graded sphere is a sphere in the gas. Every anisotropy about its CoM comes from the offset of the force centre, $r_H = -(\epsilon l^2/5) \hat{e}_3$: an antisymmetric thermophoretic torque $\gamma_0 [r_H]_x$, an antisymmetric drag coupling $-\xi_0 [r_H]_x$, and a parallel-axis rotational drag $\xi_0 z_c^2$.
- **Steady state.** The particle levitates at $G_0 = M g/\gamma_0$, independent of ϵ , heavy side down, at rest. It has no spin and no orbit, because the body is achiral and its thermophoretic torque only aligns it.
- **Dynamics.** Disturbances decay as a damped pendulum with stiffness $M g z_c$. For the defaults this is a few tens of Hz with Q of order 1. The force centre H ends exactly where it started while the CoM swings underneath it.

- **What would produce an orbit.** A steady spin needs a torque component along the spin axis, which requires chirality (for example a symmetric part of $\delta \mathbf{T}_{\text{th}}$). A circular orbit additionally needs a lateral force that rotates with the body (see [OrbitStudy.ipynb](#), Route A). Neither exists for this particle.